Standard form

1

a 2.49	**b** $20\,000\,000$	**c** $41\,300$	**d** 0.009
e $897\,000$	**f** $50\,300$	**g** 3014	**h** $8\,000\,000$
i $13\,008\,000$	**j** 0.836	**k** $7\,910\,000$	**l** 0.00714

2

a 3	**b** 5	**c** -5	**d** 10^{-7}

3

a 3×10^{2}	**b** 4.5×10^{4}	**c** 7×10^{-2}	**d** 6.23×10^{-3}
e 2×10^{3}	**f** 2×10^{-3}	**g** 1×10^{-4}	**h** 8.84×10^{5}
i 8.4626×10^{7}	**j** 6.14×10^{0}	**k** 3.47×10^{-4}	**l** 3×10^{-5}
m 7×10^{-2}	**n** 7.64×10^{-1}	**o** 2×10^{6}	**p** 1×10^{-4}

4 **D**

5

a

i Yes	**ii** No	**iii** No	**iv** Yes
v No	**vi** Yes		

b 4.5×10^{-4}

6

a

i 7.8×10^{2}	**ii** 10

b

i 4×10^{-3}	**ii** $10\,000$

c

i 9×10^{9}	**ii** $30\,000$

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7

a

i 7.27×10^{-7} ii 2.22×10^{-7}

b

i 1.25×10^{2} ii 1.25×10^{-6}

c

i 9.37×10^{-4} ii 6×10^{-9}

d

i 8.31×10^{-9} ii 5.13×10^{-9}

e

i 7.88×10^{2} ii 2.93×10^{-2}

f

i 8.99×10^{-3} ii 1.99×10^{-6}

g

i $4.7 \times 10^{9} \times 10^{8}$ ii $4.7 \times 10^{-8} \times 10^{8}$

h

i $2.12 \times 10^{7} \times 10^{2}$ ii $2.12 \times 10^{-7} \times 10^{4}$

8

a 6×10^{11} **b** 1.6×10^{-7} **c** 3×10^{3} **d** 3×10^{2}

e 4.5×10^{4} **f** 2.1×10^{10} **g** 2.4×10^{4} **h** 4.4×10^{5}

Standard form and calculators

9

a $581\,000$ **b** 0.000316 **c** $2\,500\,000$ **d** 0.00705

10

a 0.003844 **b** 0.013

11

a 1.120×10^{-15} **b** 3.342×10^{2}

12 At any particular time, approximately 2.32×10^{5} aircraft are registered as being in flight over 7.772×10^{10} square metres of air space.

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Significant figures

13
a Three **b** Four **c** Four **d** Three

14
a 5.0 **b** 0.0060 **c** $801\,600\,000$ **d** 0.0507

15
a
i 3 **ii** $422\ \text{m}^3$

b
i 2 **ii** $2.5\ \text{mL}$

16
a 2.14×10^9 **b** 1.98×10^{-5}

17 4549

18 512

19
a 3499 **b** 2500 **c** 999

20
a 1 **b** 1.09

Applications

21 The diameter of a human cell

22
a 1.332×10^9 **b** 1.332×10^{21} litres

23
a 2 million each second **b** $120\,000\,000$

24
a 1 hundred thousandth of a metre **b** $0.01\ \text{mm}$

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25

a 3×10^{-1} km/s **b** 3×10^{5} km/s **c** 1 000 000

26

a 2.5×10^{5} km^2

b 6.8×10^{7}

c 2.72×10^{2} per km^2

27

a 8×10^{5} metres per hour

b 1000

28 3300

29 135 000 kg

30 2×10^{7} g

31 7.95×10^{7}

32 2000 kg

33

a 1595 g

b 2×10^{26} atoms

34

a 2.107×10^{24} atoms

b 9 g

c 1.495×10^{-23} g

35

a 29 998 800 m

b 7×10^{8} m

36 1.4130×10^{9} days

37 1.25×10^{14}

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